

Prompt Engineering Cheat Sheet

◆ What is Prompt Engineering?

Prompt engineering is the art of designing effective instructions or questions (called "prompts") to get accurate and useful responses from AI models like ChatGPT, Claude, Gemini, etc.

◆ Common Prompt Types

Prompt Type	Description	Example
Instruction Prompt	Direct command or task	"Summarize this paragraph in one sentence."
Role Prompt	Assigns a persona or role to the AI	"You are a fitness coach. Suggest a workout plan."
Few-Shot Prompt	Gives 1–3 examples before asking for a new one	"Translate: Hello → Bonjour, Thank you → Merci..."
Chain-of-Thought	Encourages step-by-step reasoning	"Solve this problem step-by-step..."
Comparative Prompt	Compares two ideas or objects	"Compare Python and JavaScript in a table."
Creative Prompt	Asks for stories, jokes, or poems	"Write a bedtime story about a dinosaur and an alien."

◆ Prompt Structure Template

Role: You are a [expert/profession]
Task: Your task is to [goal]
Input: [Any data or question]
Instruction: [How should the AI respond?]
Constraints: [Word count, tone, format, etc.]

◆ Prompt Tuning Tips

Technique	Description
Be Specific	Vague prompts = vague outputs. Be clear and precise.
Add Context	Give relevant background or examples.
Use Constraints	Control the output length, format, or tone.
Ask for Step-by-Step	Improve logical reasoning and answers.
Limit Hallucination	Ask for citations or verify answers separately if accuracy is important.

◆ Example Comparisons

Weak Prompt:

"Tell me about AI."

Strong Prompt:

"Explain AI in simple terms to a 12-year-old in 3 paragraphs with real-life examples."

◆ Parameters (For Advanced Use)

Parameter	Description	Example Value
temperature	Controls randomness	0.2 (precise) - 0.9 (creative)
top_p	Limits vocabulary to most likely tokens	0.8–1.0

<code>max_tokens</code>	Limits output length	e.g., 200
<code>stop</code>	Ends the response at a certain word	e.g., <code>stop: ["\n\n"]</code>

◆ Fun Prompt Ideas for Practice

1. "Pretend you're Iron Man. Give life advice."
2. "Explain how a microwave works to a caveman."
3. "Create a quiz with 5 questions about planets."
4. "Write a rap song about math homework."

IMAGE PROMPT TECHNIQUE – F.R.A.M.E

- ◆ **F – Focus (Main subject)**
- ◆ **R – Rendering (Style / realism)**
- ◆ **A – Attributes (Details)**
- ◆ **M – Medium & Light (Camera, lighting)**
- ◆ **E – Environment (Background)**

Example 1: AI Engineer Working

Prompt:

Focus: A young AI engineer working on a laptop

Rendering: Ultra-realistic cinematic style

Attributes: Male, mid-20s, glasses, focused expression, casual hoodie

Medium & Light: Soft blue monitor light, shallow depth of field, 50mm lens

Environment: Modern tech workspace with code on screen, blurred city lights in background

Example 2: Student Learning Online

Prompt:

Focus: College student studying online

Rendering: Clean digital illustration style

Attributes: Female student, headphones, notebook, smiling

Medium & Light: Warm indoor lighting, soft shadows

Environment: Bedroom study setup with books, plant, and laptop on desk

Example 3: Startup Office Scene

Prompt:

Focus: Small startup team brainstorming

Rendering: Semi-realistic vector art

Attributes: Diverse team, sticky notes, whiteboard, energetic mood

Medium & Light: Bright daylight, wide-angle view

Environment: Modern open office with glass walls and city view

VIDEO PROMPT TECHNIQUE – S.A.M.E

- ♦ **S – Scene (Location & mood)**
- ♦ **A – Actor (Who is present)**
- ♦ **M – Motion (Action & camera movement)**
- ♦ **E – Edges of Time (Duration, start & end)**

Example 1: Coding Reels / Short Video

Prompt:

Scene: Night-time home workspace, minimal lighting

Actor: Young software developer typing on laptop

Motion: Camera slowly zooms toward screen, fingers typing fast

Edges of Time: 8-second clip, start wide shot, end close-up on code

Example 2: Entrepreneur Motivation Video

Prompt:

Scene: Modern office during early morning

Actor: Startup founder holding coffee, looking out window

Motion: Camera pans from coffee cup to face, slight head nod

Edges of Time: 10-second continuous shot, start side angle, end front view

Example 3: Online Teaching Video

Prompt:

Scene: Bright home studio with whiteboard

Actor: Online instructor explaining AI concepts

Motion: Hand gestures while explaining, camera fixed with slight shake

Edges of Time: 12-second clip, start medium shot, end smiling close-up

Pro Tip for Students

- **Images** → Describe visuals clearly
- **Videos** → Describe movement + time
- Always think like a **director**, not a user